Session 4:
DATA MANIPULATION



KHOA CÔNG NGHỆ THÔNG TIN
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Data Manipulation

- Computer Architecture
- Machine Language
- Program Execution
- Arithmetic/Logic Instructions
- Communicating with Other Devices
- Other Architectures



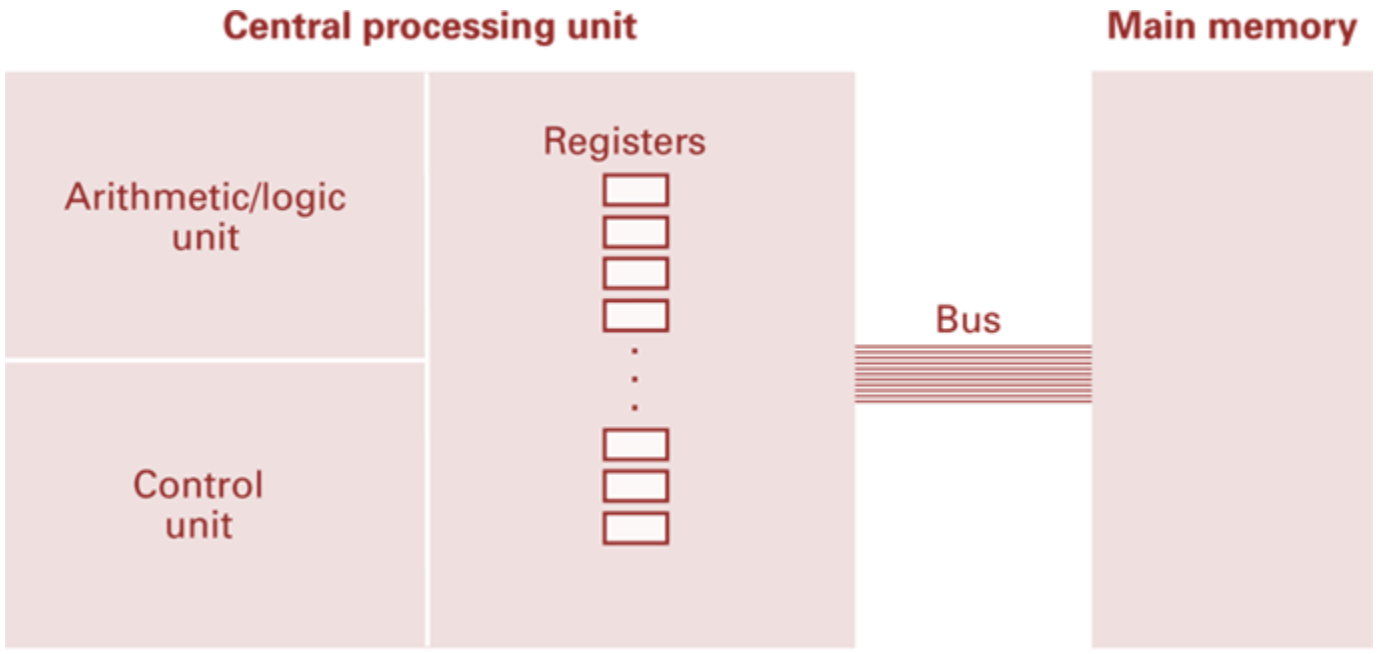
COMPUTER ARCHITECTURE



Computer Architecture

- Central Processing Unit (CPU) or processor
  - Arithmetic/Logic unit versus Control unit
  - Registers
    - General purpose
    - Special purpose
  - Cache memory
- Main memory
- Bus
- Motherboard
- I/O devices

CPU and main memory connected via a bus



Motherboards

- The motherboard is the main printed circuit board containing:
  - Buses, or electrical pathways found in a computer. Buses allow data to travel among the various components.
  - CPU, RAM,
  - Slots, heat sink/fan assembly, BIOS chip, chip set, sockets, internal and external connectors, various ports, and the embedded wires that interconnect the motherboard components.



Central Processing Unit

- Brain of the computer.
- Processor.
- The CPU executes a **program**, which is a sequence of stored **instructions**:
  - Reduced Instruction Set Computer (RISC)
  - Complex Instruction Set Computer (CISC)

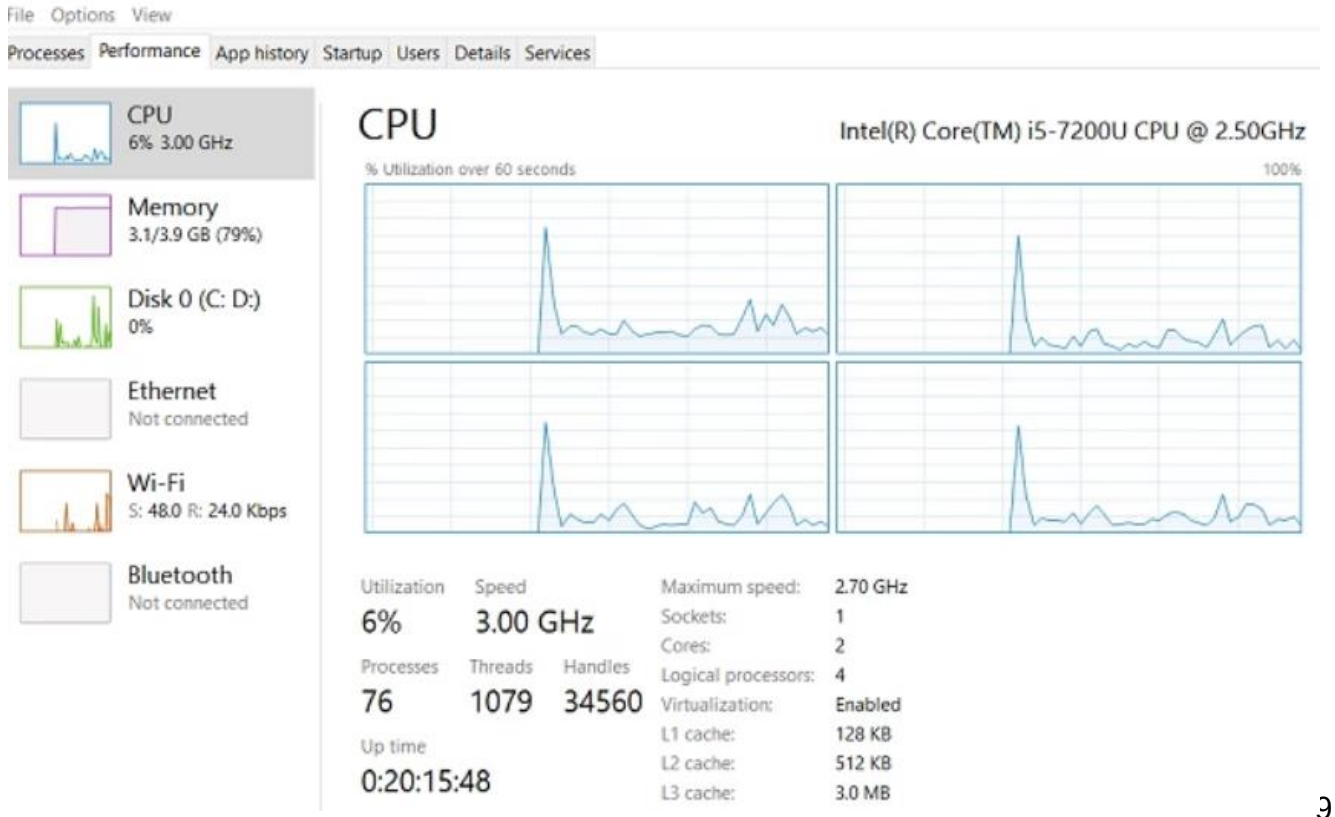


Central Processing Unit

- Some CPUs incorporate **hyperthreading** (siêu phân luồng) or **hypertransport** (siêu vận tải) to enhance the performance of the CPU.
- The amount of data that a CPU can process at one time depends on the size of the processor data bus.
  - Speed of the CPU is measured in **cycles per second** megahertz (MHz) or gigahertz (GHz).
- **Overclocking** is a technique used to make a processor work at a faster speed than its original specification.

Central Processing Unit

- The latest processor technology has resulted in CPU manufacturers finding ways to incorporate more than one CPU core onto a single chip.
  - Dual Core CPU
  - Triple Core CPU –
  - Quad Core CPU
  - Hexa-Core CPU
  - Octa-Core CPU
  - Deca-Core CPU



Arithmetic/Logic Unit

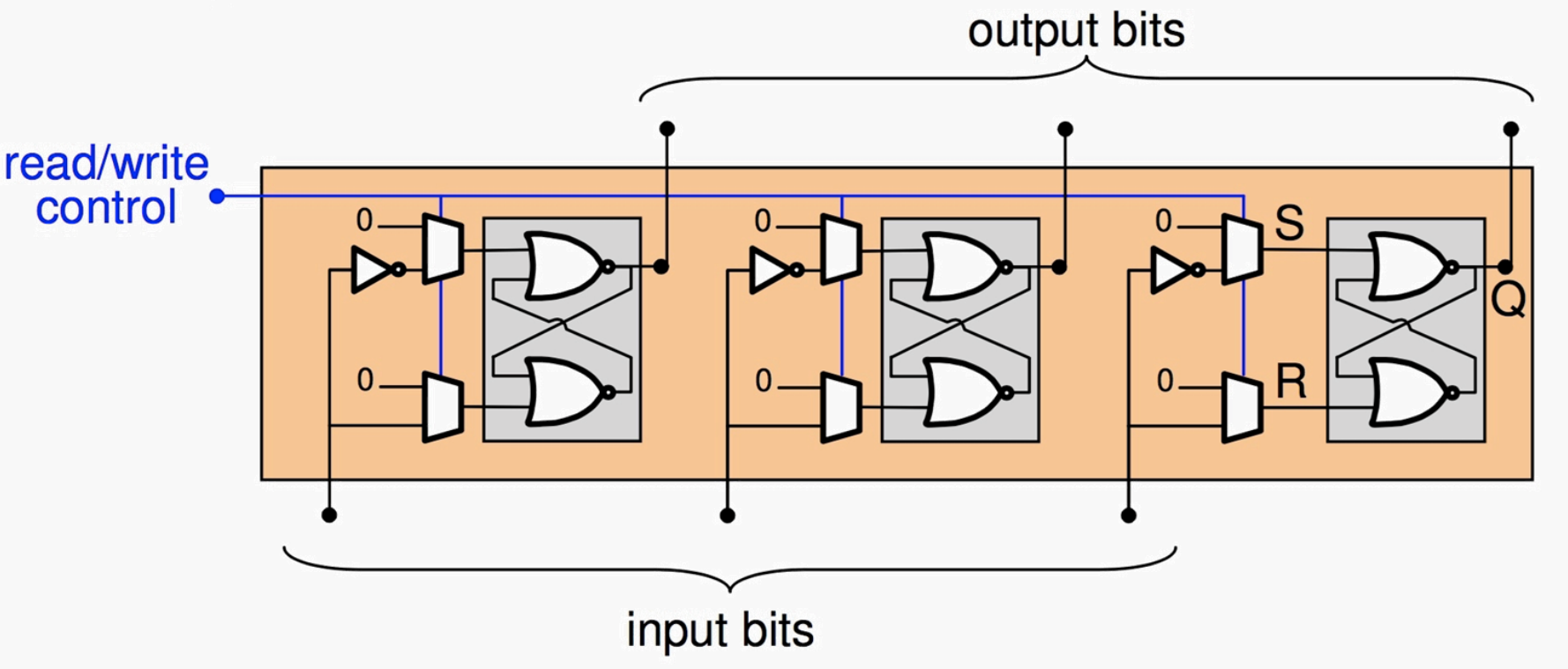
- Subsystem that performs addition, subtraction, and comparison for equality, AND, OR, NOT, XOR, …
- Components
  - Registers, interconnections between components, and the ALU circuitry
- Register
  - Storage cell that holds the operands of an arithmetic operation and holds its result
- Bus
  - Path for electrical signals

Register

- Registers are similar to RAM with following minor differences
  - They do not have a numeric memory address but are accessed by a special register designator such as A, X or R0
  - They can be accessed much more quickly than regular memory cells
  - They are not used for general purpose storage but for specific purposes such as holding the operands for an upcoming arithmetic computations.
- A typical ALU has 16, 32 or 64 registers.

Register

- Register: example of 3-bit register



Note: is denoted as , it is called a NOR gate. is a 2-to-1 multiplexor.

Source: Chun-Jen Tsai, ics12, National Chiao Tung University

Control Unit

- Responsible for interpreting the program instructions and controlling the operation
  - Tasks: fetch, decode, and execute
- Directs the operation of the other units by providing timing and control signals
- Directs the flow of data between the CPU and the other devices.
- Regulated by system clock speed



MACHINE LANGUAGE



Terminology

- Stored program
  - A program can be encoded as bit patterns and stored **in main memory**.
  - CPU can then **extract** the **instructions** and **execute** them.
  - Program to be executed can be altered easily.
  - Special form of data

Terminology

- When we ask others to do something, we use human language
  - Cumbersome and ambiguous for computers to decode and execute
- Machine instruction: An instruction (or command) encoded as a bit pattern recognizable by the CPU
- Machine language: The set of all instructions recognized by a machine
  - Concise: easy to decode
  - Precise: An instruction can be executed only in one way

Machine Language Philosophies

- Reduced Instruction Set Computing (RISC)
  - Few, simple, efficient, and fast instructions
  - Examples: PowerPC from Apple/IBM/Motorola and ARM
- Complex Instruction Set Computing (CISC)
  - Many, convenient, and powerful instructions
  - Example: Intel

Types of Machine Instruction

- **Data Transfer**
  - copy data from one location to another
- **Arithmetic/Logic**
  - use existing bit patterns to compute a new bit patterns
- **Control**
  - direct the execution of the program

Adding values stored in memory

Step 1. Get one of the values to be added from memory and place it in a register.

Step 2. Get the other value to be added from memory and place it in another register.

Step 3. Activate the addition circuitry with the registers used in Steps 1 and 2 as inputs and another register designated to hold the result.

Step 4. Store the result in memory.

Step 5. Stop.

Source: Computer Science - An Overview, 12e

Dividing values stored in memory

Step 1. LOAD a register with a value from memory.

Step 2. LOAD another register with another value from memory.

Step 3. If this second value is zero, JUMP to Step 6.

Step 4. Divide the contents of the first register by the second register and leave the result in a third register.

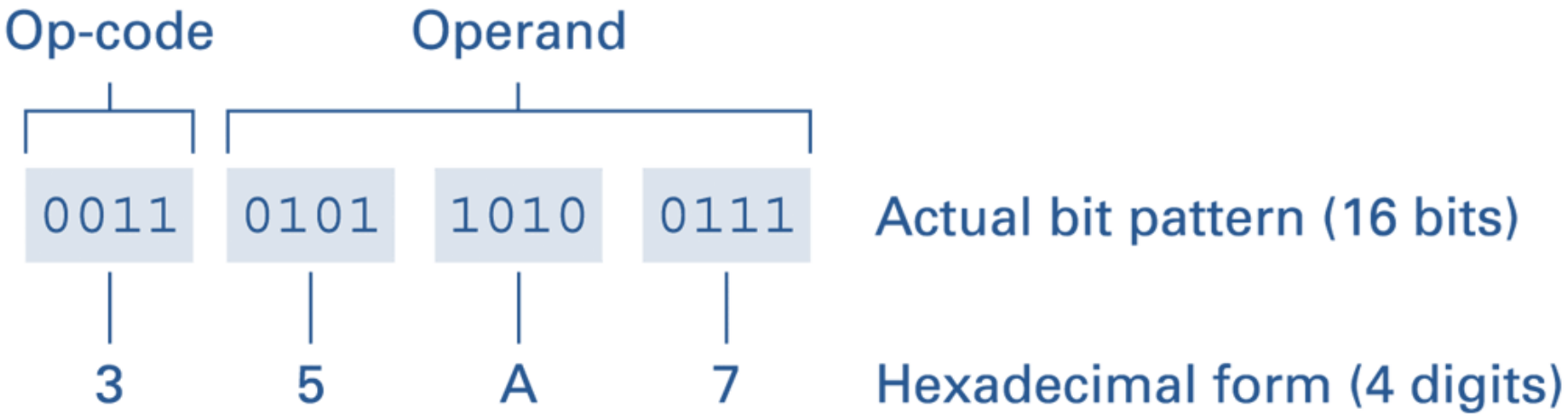
Step 5. STORE the contents of the third register in memory.

Step 6. STOP.

Parts of a Machine Instruction

- **Op-code** (mã lệnh): Specifies which operation to execute
- **Operand** (tham số): Gives more detailed information about the operation
  - Interpretation of operand varies depending on op-code

Composition of an instruction



Decoding the instruction 35A7



Op-code 3 means to store the contents of a register in a memory cell.

This part of the operand identifies the register whose contents are to be stored.

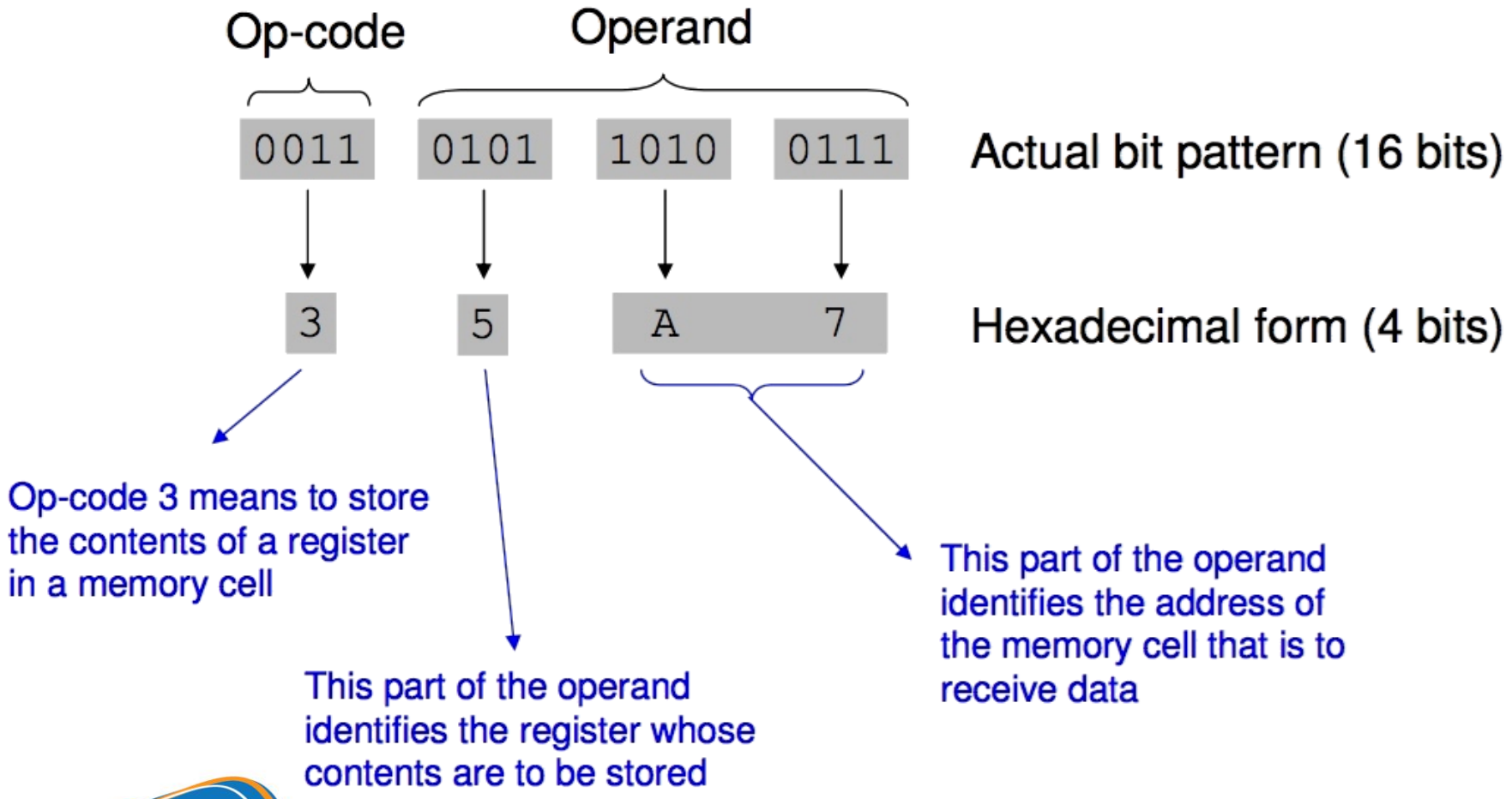
This part of the operand identifies the address of the memory cell that is to receive data.

A Simple Machine Language

Op-code	Operand	Description
1	RXY	LOAD reg. R from cell XY
2	RXY	LOAD reg. R with XY
3	RXY	STORE reg. R at XY
4	0RS	MOVE R to S
5	RST	ADD S and T into R (2's comp.)
6	RST	ADD S and T into R (floating pt.)
7	RST	OR S and T into R
8	RST	AND S and T into R
9	RST	XOR S and T into R
A	R0X	ROTATE reg. R X times
B	RXY	JUMP to XY if R = reg. 0
C	0	HALT

Example

- STORE the bit pattern found in register 5 in the memory cell whose address is A7



Source: Chun-Jen Tsai, ics12, National Chiao Tung University

A Simple Machine Language

- **1**4A3: Load
  - The contents of the **memory cell** located at address **A3** to be placed in **register** 4.
- **2**0A3: Load
  - The value **A3** to be placed in **register** 0.
- **3**5B1: Store
  - The contents of register 5 to be placed in the **memory cell** whose address is **B1**.
- **4**0A4: Move
  - The contents of **register A** to be copied into **register 4**.

A Simple Machine Language

- 5726: Add
  - The binary values in **registers** 2 and 6 to be added and the sum placed in **register** 7.
- 634E: Add
  - The values in registers 4 and E to be added as **floating-point** values and the result to be placed in register 3.

A Simple Machine Language

- 7CB4: Or
  - The result of **OR**ing the content of **registers** B and 4 to be placed in **register** C.
- 8045: And
  - The result of **AND**ing the content of **registers** 4 and 5 to be placed in **register** 0.
- 95F3: Xor
  - The result of **XOR**ing the contents of **registers** F and 3 to be placed in **register** 5.
- A403: Rotate
  - The content of **registers** 4 to be rotated 3 bits to the right in a circular fashion.

A Simple Machine Language

- B43C:
  - Compare the contents of **register 4** with the contents of **register** 0.
  - If equal, the pattern **3C** would be placed in the program counter so that the next execution executed would be the one located at that memory address.
  - Otherwise, nothing would be done.
- C000:
  - Stop program execution.

An encoded version of the instructions

Encoded instructions	Translation
156C	Load register 5 with the bit pattern found in the memory cell at address 6C.
166D	Load register 6 with the bit pattern found in the memory cell at address 6D.
5056	Add the contents of register 5 and 6 as though they were two's complement representation and leave the result in register 0.
306E	Store the contents of register 0 in the memory cell at address 6E.
C000	Halt.

Quiz

- The following are instructions written in the above simple machine language.
  - 368A
  - BADE
  - 803C
  - 40F4

Quiz

- What is the difference between the instructions **15AB** and **25AB** in the simple machine language?

Quiz

- Here are some instructions in English. Translate each of them into the simple machine language.
  - LOAD register number 3 with the hexadecimal 56.
  - ROTATE register number 5 three bits to the right.
  - AND the contents of register A with the contents of register 5 and leave the result in register 0.

Quiz

- Here are some instructions in English. Translate each of them into the simple machine language.
  - LOAD register number 3 with the hexadecimal 56.

2356

  - ROTATE register number 5 three bits to the right.

A503

  - AND the contents of register A with the contents of register 5 and leave the result in register 0.

80A5



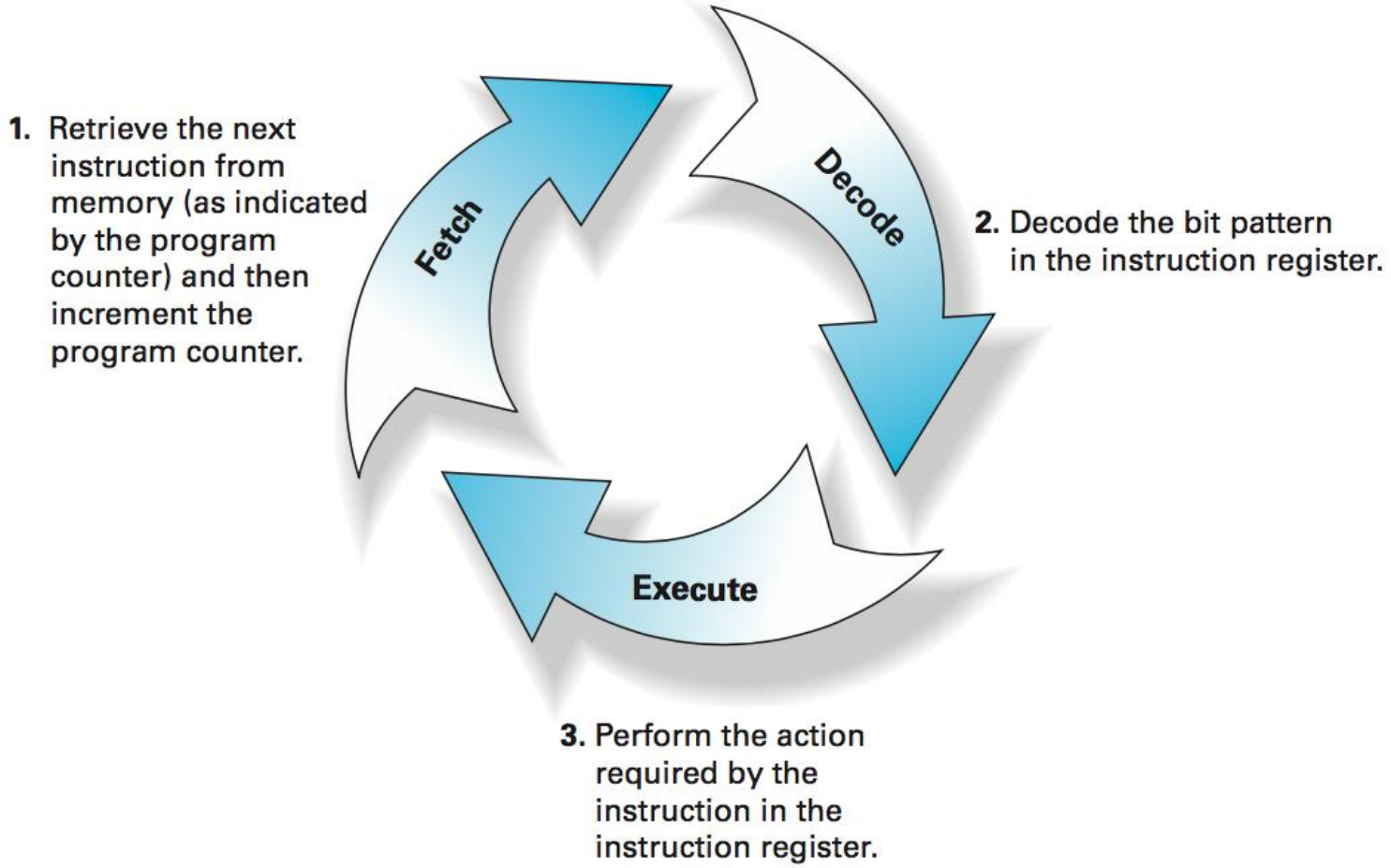
PROGRAM EXECUTION



Program Execution

- Controlled by two special-purpose registers
  - Program counter (bộ đếm): address of next instruction
  - Instruction register (chỉ lệnh): current instruction to be executed
- Machine Cycle
  - Fetch
  - Decode
  - Execute

Machine cycle



Decoding the instruction B258

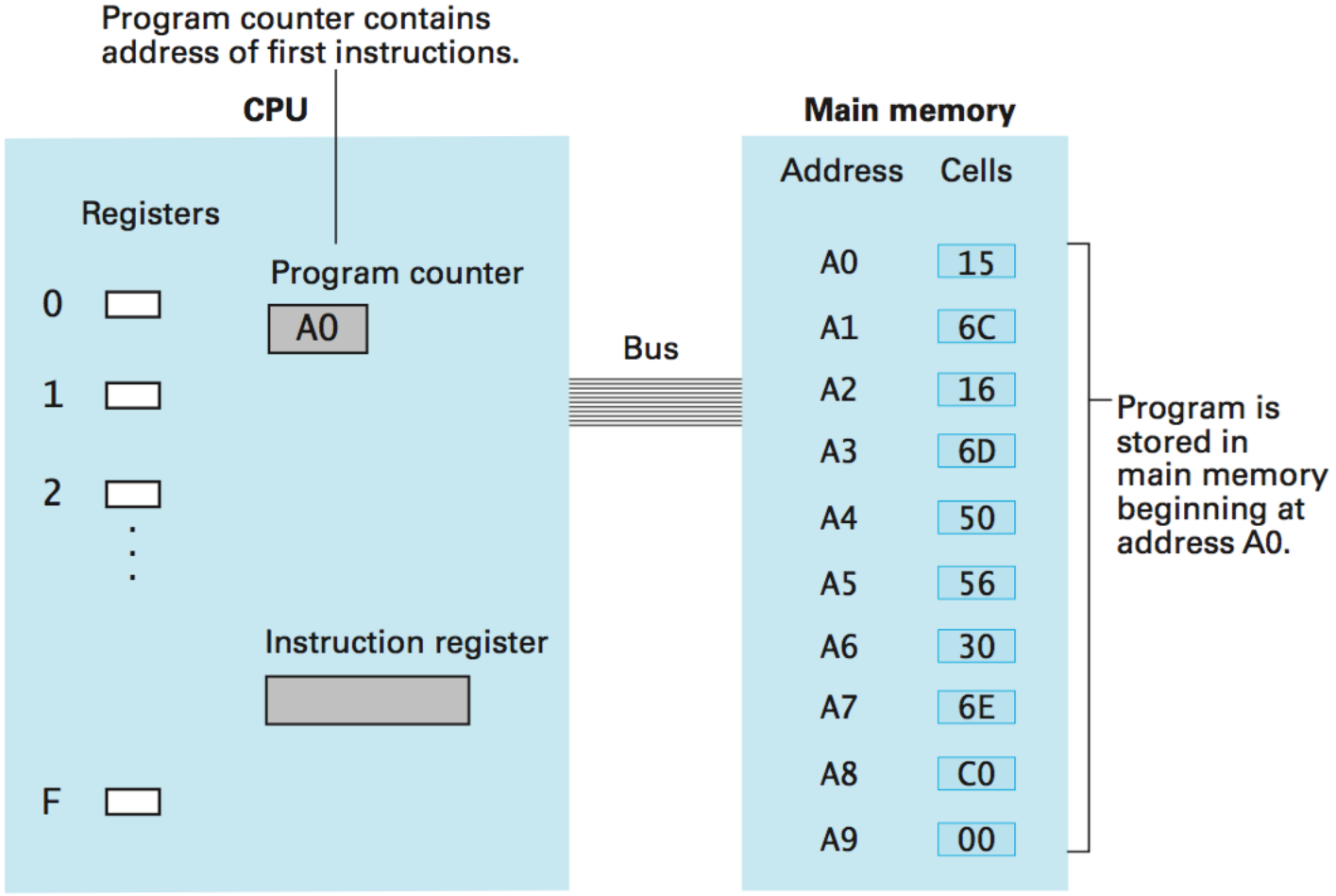


Op-code B means to change the value of the program counter if the contents of the indicated register is the same as that in register 0.

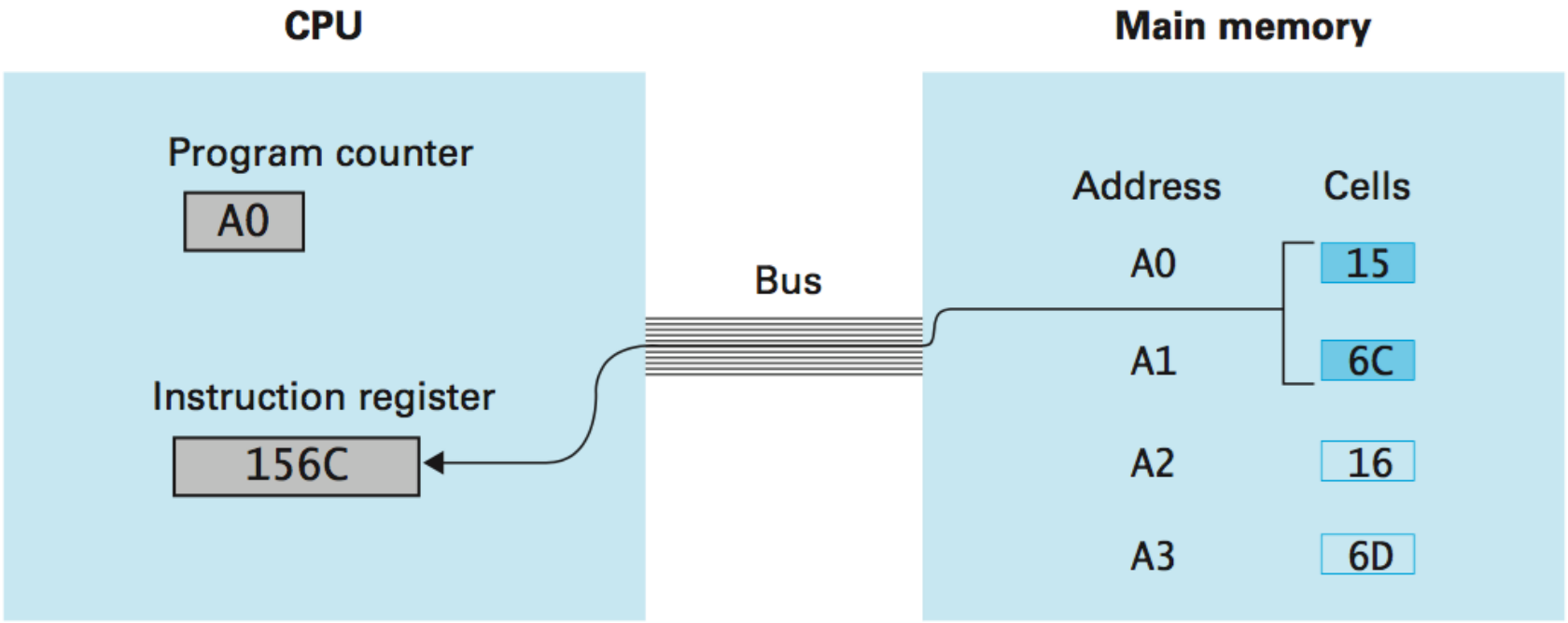
This part of the operand identifies the register to be compared to register 0.

This part of the operand is the address to be placed in the program counter.

Program stored in main memory



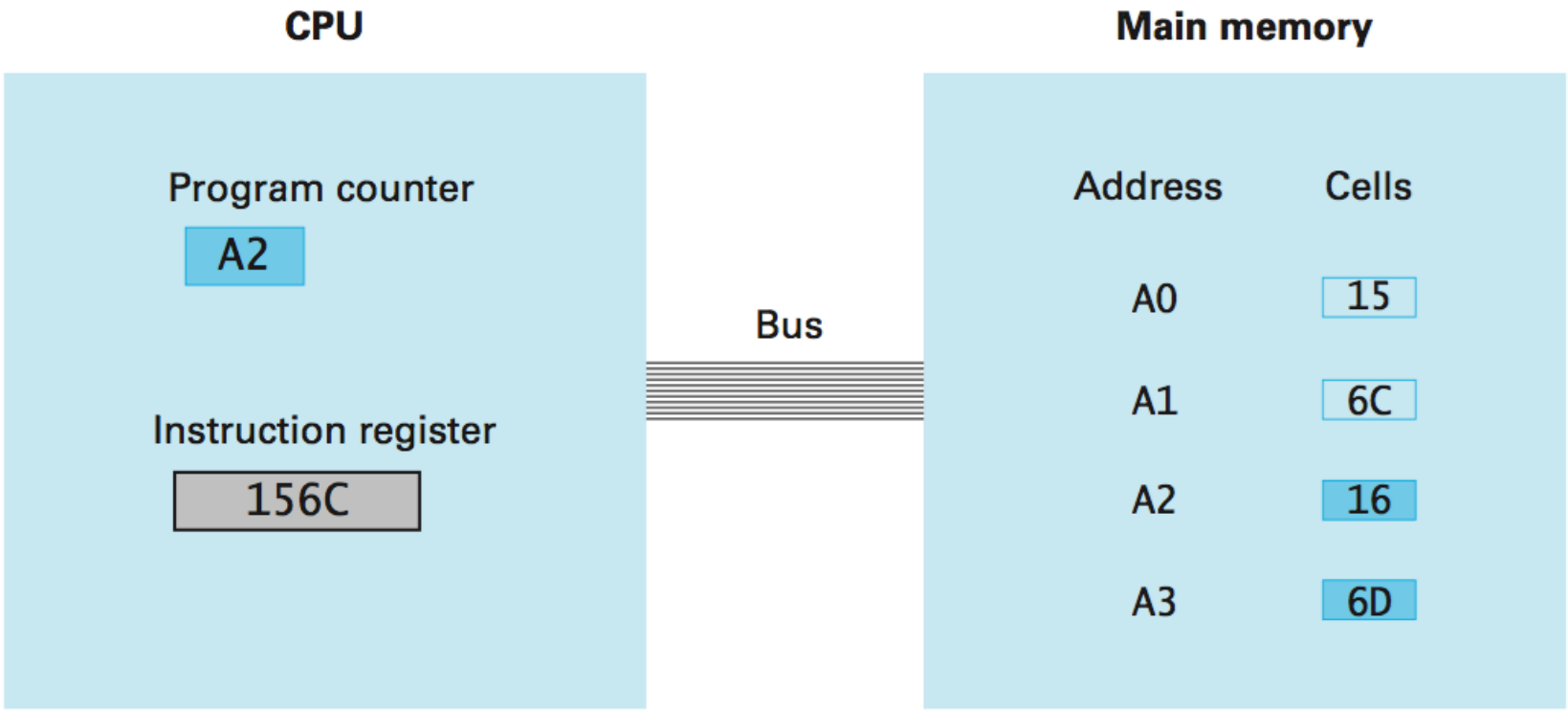
The fetch step of the machine cycle



a. At the beginning of the fetch step the instruction starting at address A0 is retrieved from memory and placed in the instruction register.

Source: Computer Science - An Overview, 12e

The fetch step of the machine cycle



b. Then the program counter is incremented so that it points to the next instruction.

ARITHMETIC/LOGIC INSTRUCTIONS

Arithmetic/Logic Operations

- Logic: AND, OR, XOR
  - Masking
- Rotate and Shift:
  - circular shift (rotation)
  - logical shift
  - arithmetic shift
- Arithmetic: add, subtract, multiply, divide
  - Precise action depends on how the values are encoded (two's complement versus floating-point).

Masking

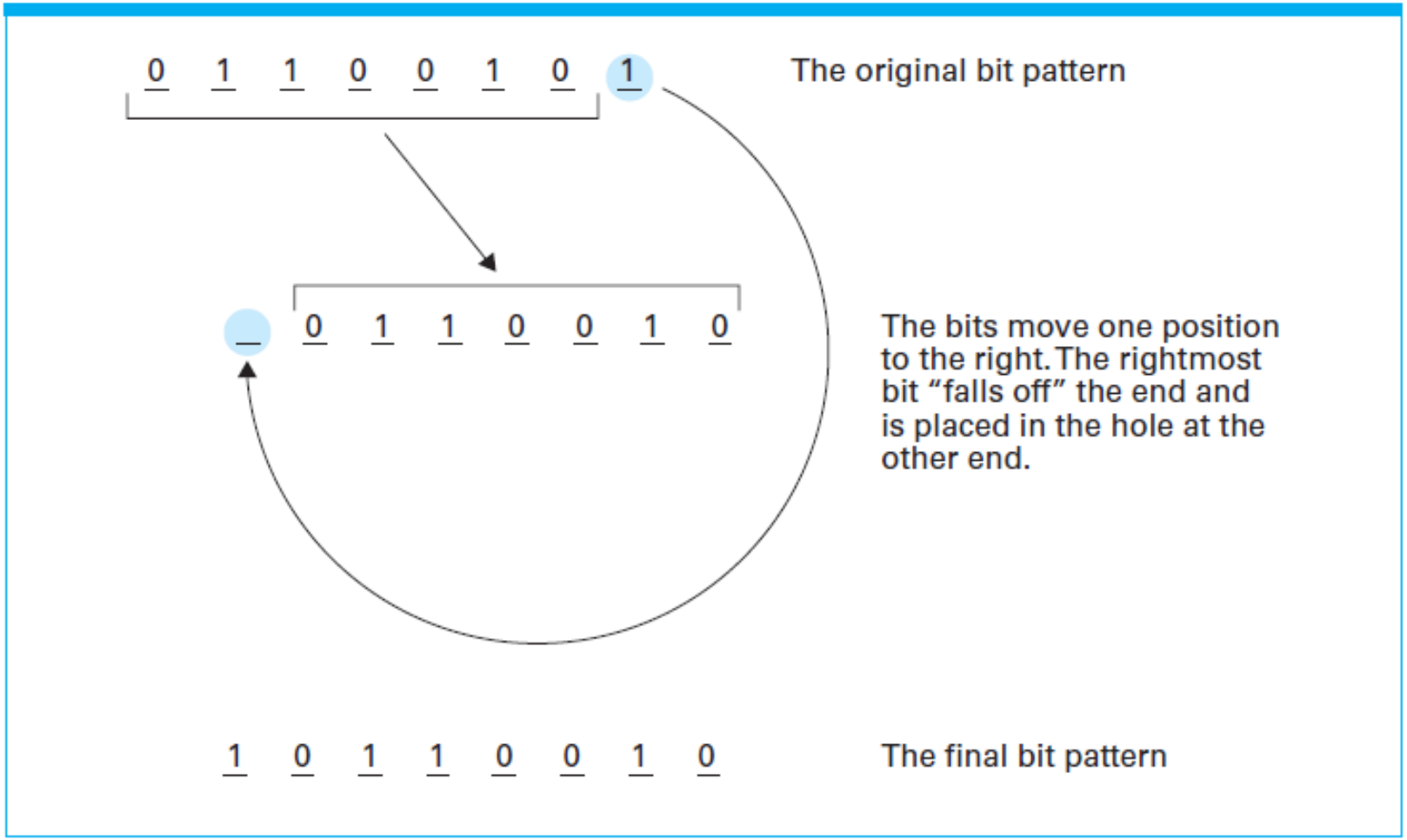
```
  00001111
AND 10101010
  00001010
```

```
  11110000
OR  10101010
  11111010
```

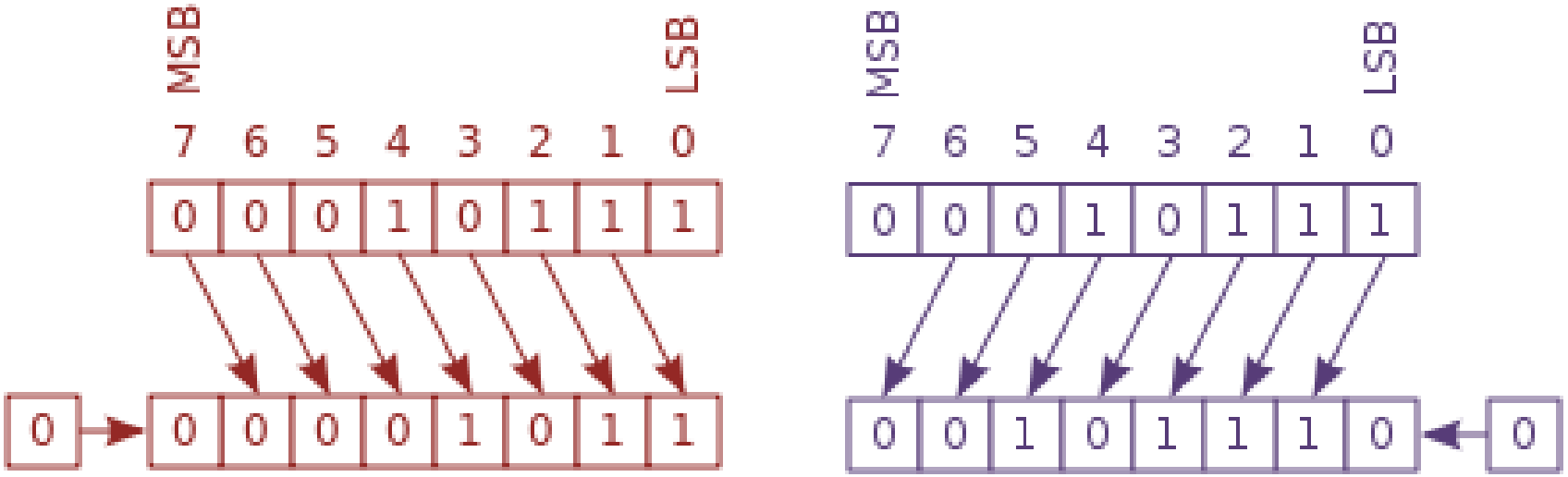
```
  11111111
XOR 10101010
  01010101
```

Circular shift

Rotating the bit pattern 65 (hexadecimal) one bit to the right



Logical shift



Quiz

- What logical operation together with what mask can you use to change ASCII codes of lower characters to uppercase? What about uppercase to lowercase?

Symbol	ASCII	Hex	Symbol	ASCII	Hex
>	00111110	3E	^	01011110	5E
?	00111111	3F	_	01011111	5F
@	01000000	40	`	01100000	60
A	01000001	41	a	01100001	61
B	01000010	42	b	01100010	62
C	01000011	43	c	01100011	63
D	01000100	44	d	01100100	64
E	01000101	45	e	01100101	65
F	01000110	46	f	01100110	66
G	01000111	47	g	01100111	67
H	01001000	48	h	01101000	68
I	01001001	49	i	01101001	69
J	01001010	4A	j	01101010	6A
K	01001011	4B	k	01101011	6B
L	01001100	4C	l	01101100	6C

Quiz

- Identify both the mask and the logical operation needed to accomplish each of the following objectives:
- Put 1s in the upper 4 bits of an 8-bit pattern without disturbing the other bits.
- Complement the most significant bit of an 8-bit pattern without changing the other bits.
- Complement a pattern of 8 bits.
- Put a 0 in the least significant bit of an 8-bit pattern without disturbing the other bits.

Quiz

- Identify both the mask and the logical operation needed to accomplish each of the following objectives:
- Put 1s in the upper 4 bits of an 8-bit pattern without disturbing the other bits.

  OR 1111 0000
- Complement the most significant bit of an 8-bit pattern without changing the other bits.

  XOR 1000 0000
- Complement a pattern of 8 bits.

  XOR 1111 1111
- Put a 0 in the least significant bit of an 8-bit pattern without disturbing the other bits.

  AND 1111 1110

Quiz

- Identify both the mask and the logical operation needed to accomplish each of the following objectives:
- Put 1s in all but the most significant bit of an 8-bit pattern without disturbing the most significant bit.
- Filter out all of the green color component from an RGB bitmap image pixel in which the middle 8 bits of a 24-bit pattern store the green information.
- Invert all of the bits in a 24-bit RGB bitmap
- Set all the bits in a 24-bit RGB bitmap pixel to 1, indicating the color “white”.

Quiz

- Identify a logical operation (along with a corresponding mask) that, when applied to an input string of 8 bits, produces an output string of all 0s if and only ifs the input string is **1000 0001**.

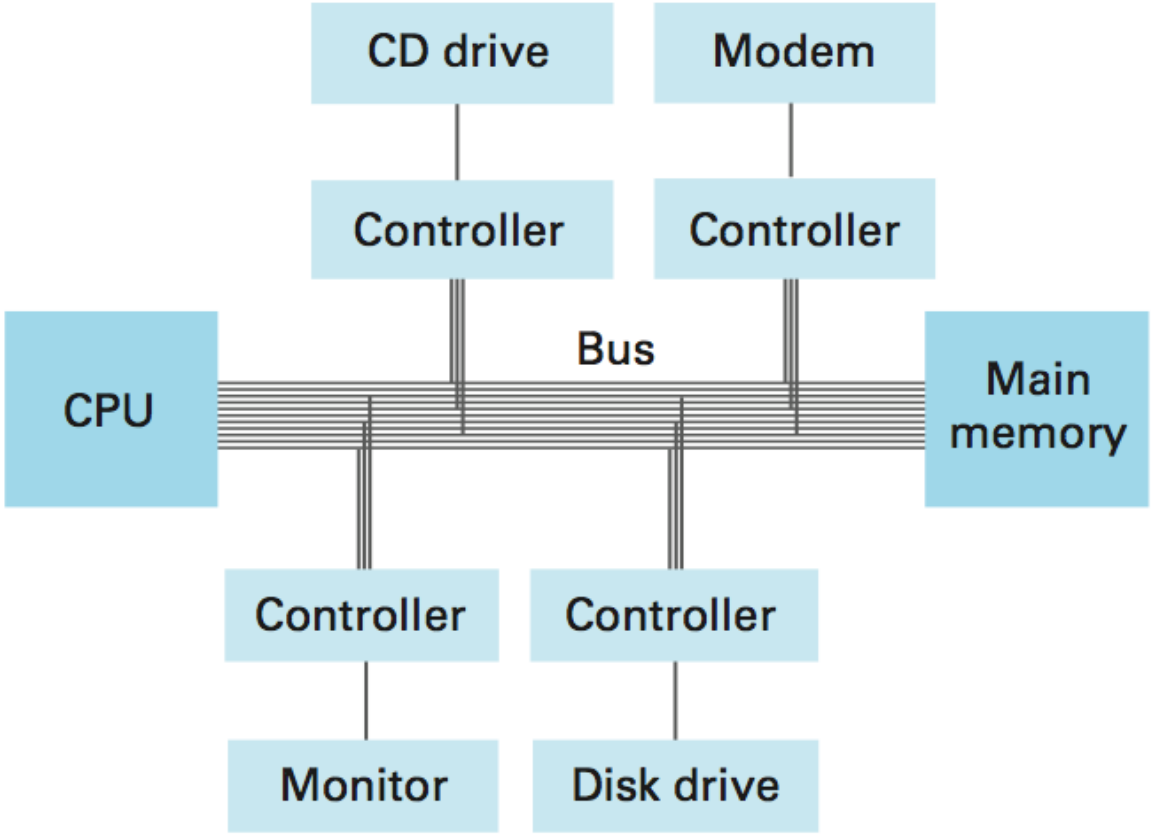


COMMUNICATING WITH OTHER DEVICES

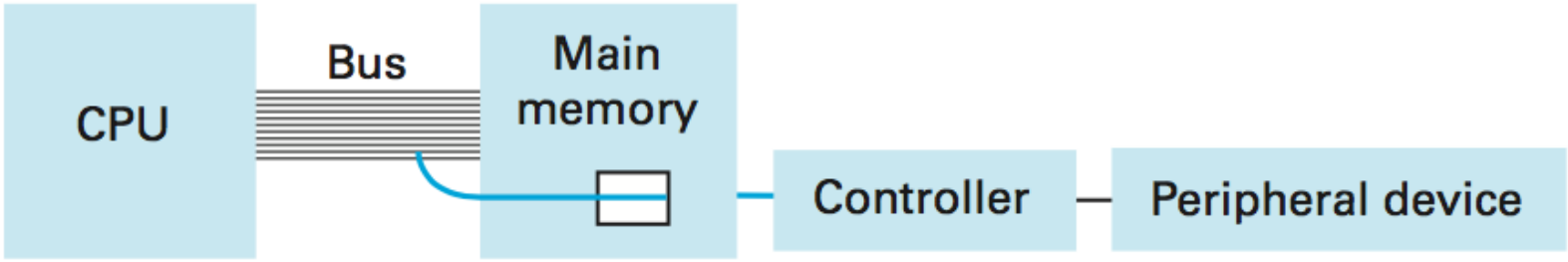
Communicating with Other Devices

- Controller: An intermediary apparatus that handles communication between the computer and a device
  - Specialized controllers for each type of device
  - General purpose controllers (USB and FireWire)
- Port: The point at which a device connects to a computer
- Memory-mapped I/O: CPU communicates with peripheral devices as though they were memory cells

Controllers attached to a machine's bus



A conceptual representation of memory-mapped I/O



Communicating with Other Devices

- Direct memory access (DMA):
  - Main memory access by a controller over the bus
- Von Neumann Bottleneck:
  - Insufficient bus speed impedes performance
- Handshaking:
  - The process of coordinating the transfer of data between components

Communicating with Other Devices

- Parallel Communication:
  - Several communication paths transfer bits simultaneously.
- Serial Communication:
  - Bits are transferred one after the other over a single communication path.

Data Communication Rates

- Measurement units
  - Bps: Bits per second
  - Kbps: Kilo-bps (1,000 bps)
  - Mbps: Mega-bps (1,000,000 bps)
  - Gbps: Giga-bps (1,000,000,000 bps)

- Bandwidth: Maximum available rate

Quiz

- Assume that the machine (using the simple machine language as described) uses memory-mapped I/O and that the address B5 is the location within the printer port to which data to be printed should be sent.

a. If register 7 contains the ASCII code for the letter A, what machine language instruction should be used to cause that letter to be printed at the printer?

b. If the machine executes a million instructions per second, how many times can this character be sent to the printer in one second?

Quiz

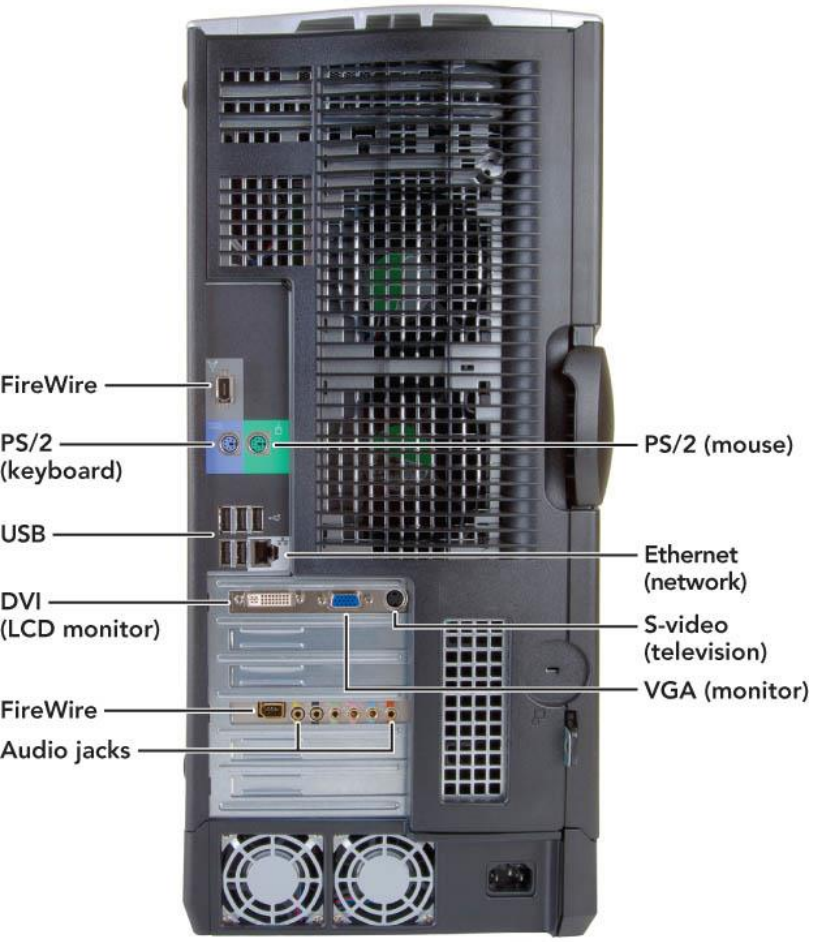
- Assume that the machine (using the simple machine language as described) uses memory-mapped I/O and that the address B5 is the location within the printer port to which data to be printed should be sent.

c. If the printer is capable of printing five traditional pages of text per minute, will it be able to keep up with the characters being sent to it in (b)?

Quiz

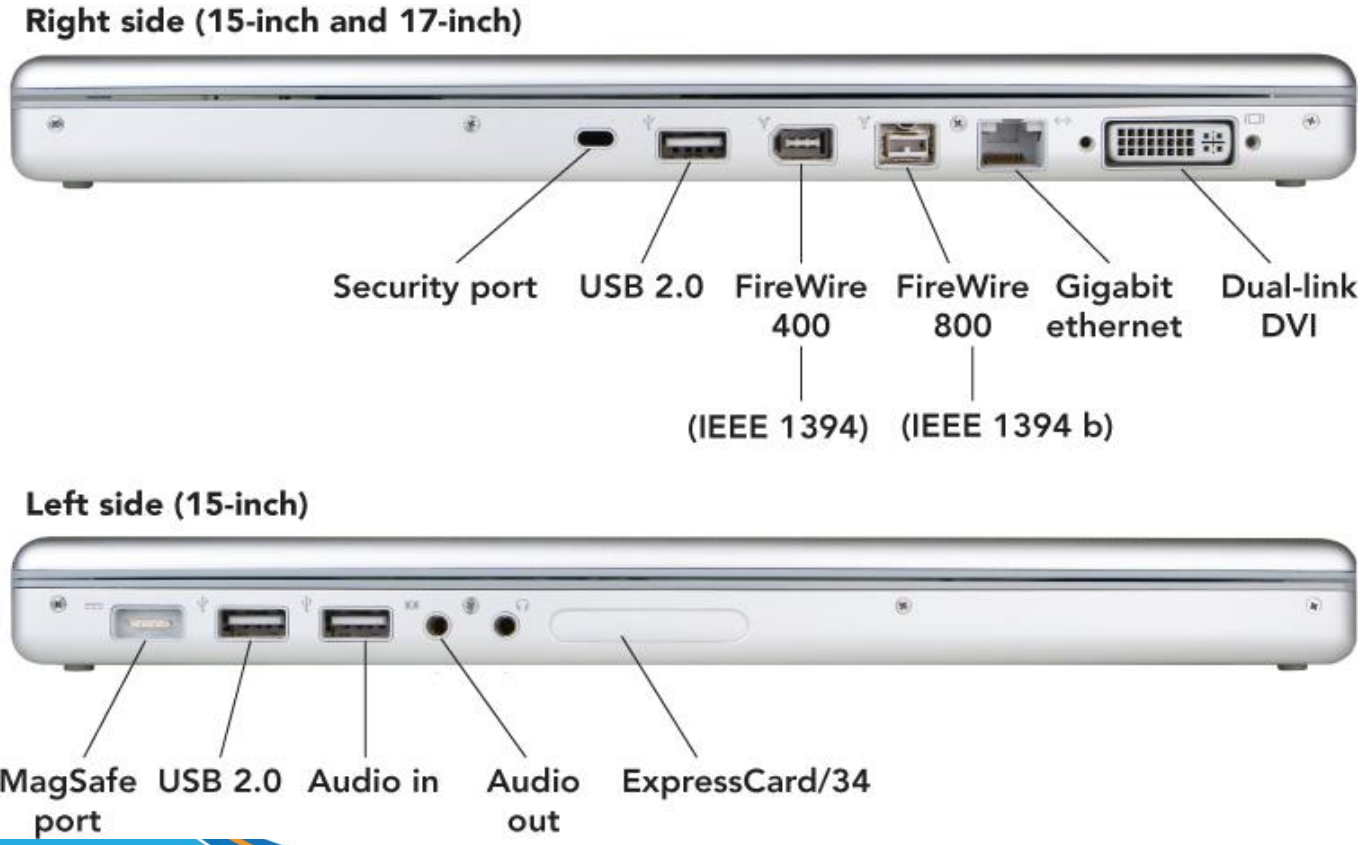
- Estimate how long it would take to transfer a 250-page novel (about 500,000 characters) encoded in 16-bit Unicode characters at a transfer rate of 54Mbps.

What's on the Outside of the Box?



What's on the Outside of the Box?

- Connectors on a notebook may vary



Universal serial bus

- USB (universal serial bus) ports
  - Connects up to 127 peripheral devices
  - USB 2.0 (high-speed USB)—fully compatible with USB 1.1 products, cables, and connectors
  - Designed to replace older parallel and serial ports
  - Connects a variety of devices to the computer, including:
    - Keyboards
    - Mice
    - Printers
    - Digital cameras



Universal serial bus

- USB 2.0
  - Uses an external bus
  - Supports data transfer rates of 480 Mbps between the computer and the peripheral device
  - Supports hot swapping—ability to connect and disconnect devices without shutting down the computer
  - Plug-and-play (PnP)—allows computers to automatically detect the device when you plug it in
- USB 3.0 (2008): 5Gbps
- USB 3.1 (2013): 10Gbps
- USB hub
  - Device that plugs into existing USB port
  - Contains four or more additional ports

FireWire

- FireWire (1395 ports)
  - Created by Apple in 1995
  - IEEE 1394 Higher Performance Serial Bus, also known as Sony i.Link
  - Offers high-speed connections for dozens of peripheral devices (up to 63)
  - Enables hot swapping and PnP
  - Data transfer rates of FireWire
    - FireWire 400—400 Mbps
    - FireWire 800—800 Mbps
    - Declared “dead” by Steve Jobs.

Video connectors

- VGA (video graphics array)
  - 15-pin male connector—works with standard monitor cables
  - Transmits analog video signals
  - Used for legacy technology cathode ray (CRT) monitors
- DVI (Digital visual Interface) port—lets LCD monitors use digital signals
- Onboard video—video circuitry built into the motherboard where the video connector is on the back of the system unit case

HDMI

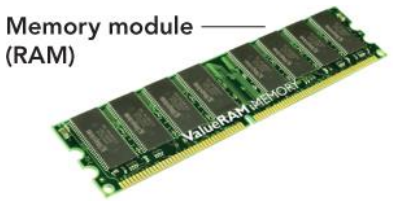
- High-Definition Multimedia Interface
  - A video/audio interface for transmitting uncompressed video data, and compressed/uncompressed audio data.
  - Digital replacement for analog video standards.
  - Designed: 2002 (7 companies).
  - HDMI 2.1: 48Gbps.
  - Newer version: 3D, Ethernet data connection, Consumer Electronic Control (CEC) extensions.

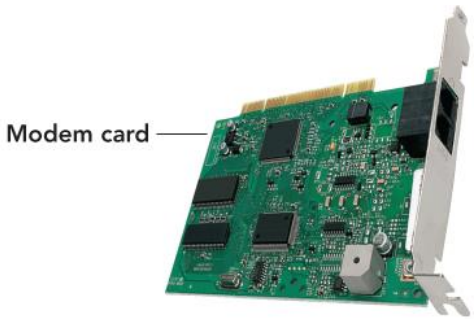




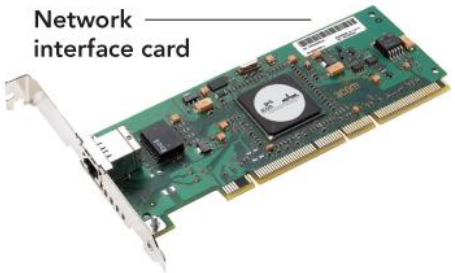
Additional connectors

- Telephone
- Network
- PC card slot
  - PC card
  - ExpressCard
- Sound card
- Game card
- TV/sound capture board







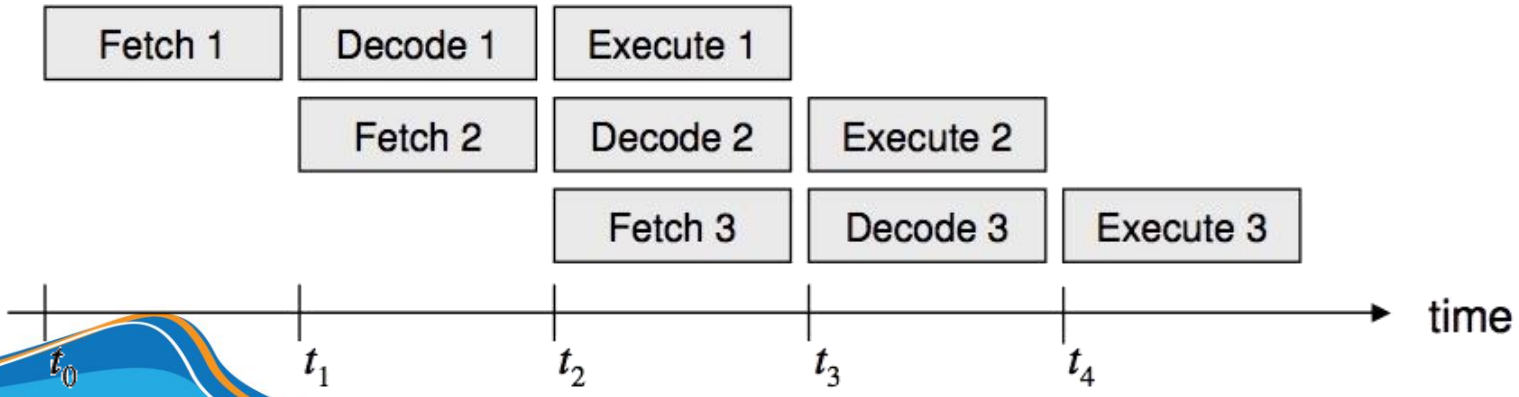




OTHER ARCHITECHTURES

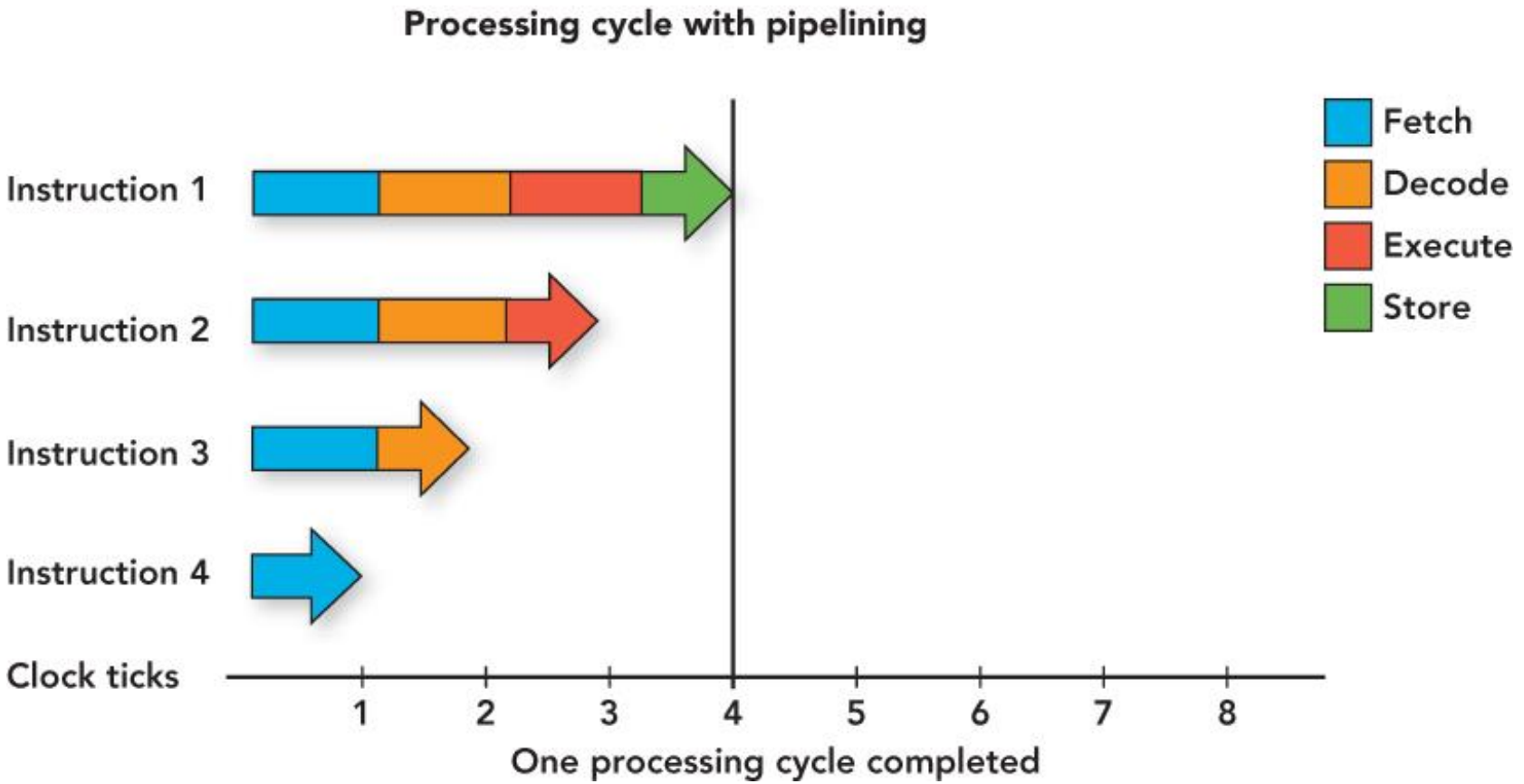
Other Architectures

- Technologies to increase throughput:
  - Pipelining: Overlap steps of the machine cycle
  - Parallel Processing: Use multiple processors simultaneously
    - SISD: No parallel processing
    - MIMD: Different programs, different data
    - SIMD: Same program, different data



Source: Chun-Jen Tsai, ics12, National Chiao Tung University

Pipelining


Processing cycle with pipelining
Fetch
Decode
Execute
Store
Instruction 1
Instruction 2
Instruction 3
Instruction 4
Clock ticks
1
2
3
4
5
6
7
8
One processing cycle completed

Parallel processing

- Method where more than one processor performs at the same time—faster processing

